Figure 3: *Visualization of personality vector clusters in a 2-dimensional space using t-SNE. Each point represents a personality trait, with the trait name as a label. Colors indicate distinct clusters of related traits.*

This grouping highlights the need to understand and address potential biases within the model's internal representations.

Further analysis of these clusters and their relationships with linguistic features and psychological constructs could provide a deeper understanding of the mechanisms by which LLMs represent and utilize personality information during language generation.

2.5 Quantifying Information with Principal Component Analysis

While dimensionality reduction techniques like PCA offer valuable insights into the structure of high-dimensional data, it's crucial to assess the amount of information preserved during this process. This section focuses on analyzing the error reduction achieved through PCA, identifying the most informative personality traits within the principal component space.

Cluster Number	Personalities
Cluster 1	depressive, nihilistic, solipsistic
Cluster 2	arrogant, biased, blunt, close-minded, confrontational, egotistical, hostile, impatient, intolerant, jerk, narcissistic, self-centered, stubborn, vindictive
Cluster 3	anxious, fragile, humble, indecisive, modest, passive, shy, submissive, timid
Cluster 4	aloof, apathetic, indifferent, neglectful, uninterested, unsympathetic
Cluster 5	adventurous, adventurous spirit, energetic, extroverted, fun-loving, homebody, humorous, outgoing, sociable
Cluster 6	catatonic, introverted, loner, reserved, schizoid, solitary
Cluster 7	dependent, emotional, emotional thinker, intuitive, neurotic, schizotypal, sensitive
Cluster 8	cannibalistic, corrupt, greedy, lustful, masochistic, molestful, murderous, pedophilic, psychopathic, sadistic, sociopathic, torturous, vengeful, violent
Cluster 9	analytical, data-driven, logical, practical, practical thinker, rational, realistic, scientific
Cluster 10	easygoing, flexible, inquisitive, laid-back, open-minded, tolerant, yielding
Cluster 11	amiable, artistic, charismatic, content, friendly, generous, idealistic, optimistic
Cluster 12	critical, dour, paranoid, pessimistic, pessimistic realist, skeptical, stingy
Cluster 13	autistic, cautious, detail-oriented, focused, methodical, obsessive-compulsive, organized, perfectionist, planner, rigid thinker
Cluster 14	fanatical, histrionic, passionate, turbulent, zealous
Cluster 15	big-picture, creative, creative thinker, dreamer, innovative, innovative thinker, practical dreamer, resourceful, strategic thinker, visionary, visionary pragmatist
Cluster 16	calm, diplomatic, forgiving, mentor-like, nurturing, patient, supportive
Cluster 17	altruistic, compassionate, empathetic, sympathetic
Cluster 18	disorganized, distracted, flighty, irresponsible, negligent, spontaneous, unpredictable, unreliable, unsystematic
Cluster 19	ambitious, assertive, autocratic leader, bold, competitive, confident, deceptive, determined, dishonest, independent, machiavellian, manipulative, resilient, showy, tenacious
Cluster 20	ambivert, conventional, cooperative leader, cooperative, diplomatic negotiator, ethical, fair-minded, grounded, honest, loyal, optimistic realist, persuasive, reliable, responsible, serious, sincere, traditional, trustworthy, utilitarian, vigilant

Table 3: *Cluster descriptions with associated personality traits.*

2.5.1 PCA Error Reduction and Optimal Component Selection

A key aspect of PCA is determining the optimal number of principal components (PCs) to retain for subsequent analysis. This decision involves balancing dimensionality reduction with information loss. To guide this selection, we examined the error reduction achieved with an increasing number of PCs. Specifically, we calculated the mean squared error (MSE) between the original, high-

dimensional personality vectors and the reconstructed vectors obtained using a varying number of PCs.

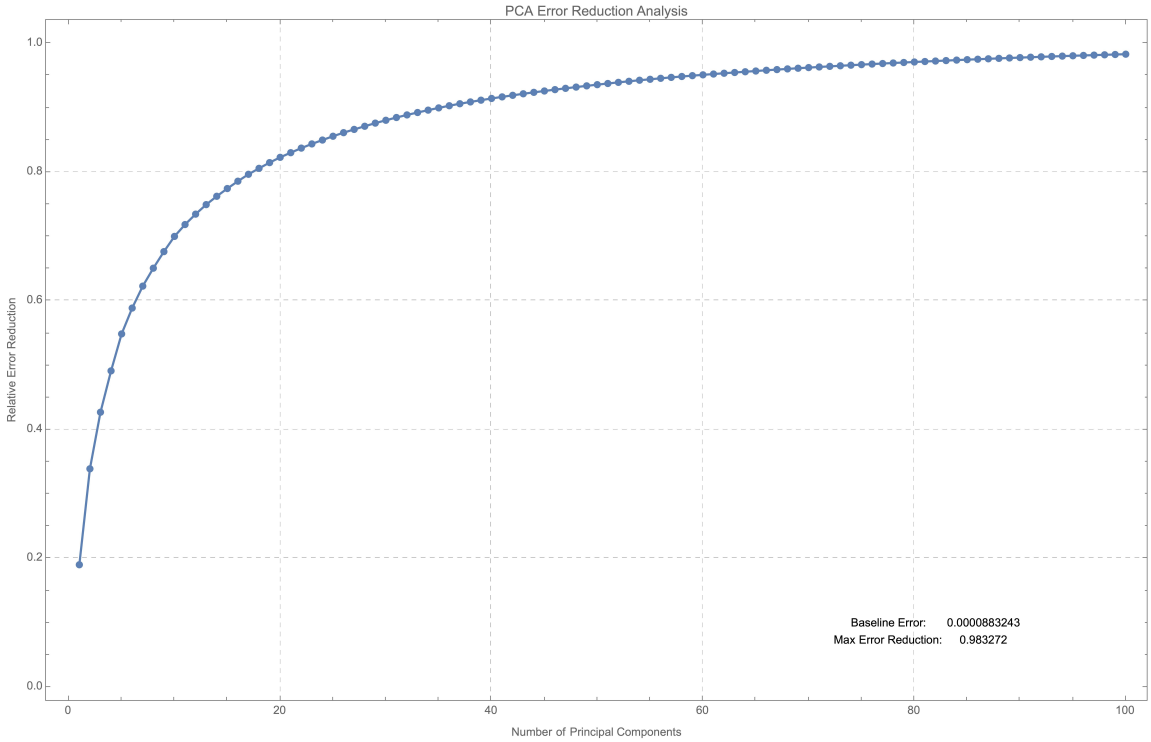

Figure 4: *Relative error reduction as a function of the number of principal components used for data reconstruction. The plot illustrates the diminishing returns in error reduction as more components are added.*

Figure 4 illustrates the relative error reduction as a function of the number of PCs. The plot clearly shows the diminishing returns in error reduction as more PCs are incorporated. This information allows us to make an informed decision regarding the optimal number of components to retain, balancing complexity reduction with information preservation for subsequent analyses.

2.5.2 Identifying Influential Personalities for Reconstruction

Beyond error reduction, we aimed to identify the most influential personality traits – those that best capture the variance within the data – when reconstructing the personality vector space from a reduced set of principal components.

To achieve this, we first standardized the personality vectors to ensure comparability across different traits. Then, using a brute-force approach, we iteratively constructed sets of basis vectors for reconstruction, starting with the single most informative personality and progressively adding